The Ethical Implications of AI in Cybersecurity Analysis

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Introduction

As a computer science/cybersecurity major who has chosen the path of becoming a cybersecurity analyst. Cybersecurity analysts play a big role in protecting digital systems from things like data breaches, cybercrime and malicious attacks. Their role is to ensure the safety of the integrity of the information systems. Know in recent years. AI has begun to become bigger than any one thought it would be and it has shaped the field of cybersecurity. AI power systems are capable of detecting vulnerabilities, anomalies, and automating the response to cyber threats with remarkable speed. While this new technology may offer new promising benefits it also raises serious ethical concerns. They would include fairness, accountability, and privacy. This article will look at how AI impacts cybersecurity, the ethical implications of its uses, and how these subjects are viewed from the viewpoints of Christian and ethical principles as well as professional computer science standards of ethics such those published by IEEE.

AI Impact on Cybersecurity

In modern cybersecurity, AI has become a crucial technological advancement. Systems that demonstrate cognitive abilities similar to those of humans, including self-correction, learning, and reasoning, are referred to as artificial intelligence, according to IBM. But in cybersecurity, the same abilities are applied to enhance the capacity to recognize and address risks. I am going to be talking in depth about what AI looks like in certain cybersecurity tastes. AI is starting to play a role in predictive defenses. When it comes to defense AI systems they can forecast vulnerability before attackers exploit them. Instead of strengthening defenses reactively, security professionals may do it pro actively thanks to this predictive power. First, let's examine danger detection and prevention. AI is also replacing it. AI can swiftly examine enormous volumes of data to find trends linked to the possible cyberattack. In order to identify abnormalities and notify analysts before damage is done, machine learning models are created. When you really think about it is so frustrating how AI is taking over because so many people are losing their jobs due to AI talking over and with me going into the cybersecurity analyst field where they do have that AI in place to basically do our jobs for us is very concerning. I do believe that AI can be a very beneficial thing but I don't think we should use it all the time. Sometimes AI could be wrong and it could cause more problems and then we have to be the one to clear it up. Because if AI is so humanized, is it really smarter than a human? I can hold a lot of data for another topic. Other things that AI effects in the cybersecurity field are automatic routines. AI is a driven automatic tool such as SOAR which is a security orchestrated automation and response service which can handle time sensitive tasks and free some like me who want to be a cybersecurity analyst to focus on higher taste and strategy and analysis. Yes AI may help us do our work faster but we always have AI in place. Will there be a place to learn or even improve in the field if AI is doing the little task that we are supposed to be doing? Some issues that I find with AI being in the cybersecurity field is that there is no privacy AI system that can collect and analyze personal data which is not a good thing because it could lead to surveillance and loss of privacy. Ai taking over would be unfair and it could be biased to certain groups or users. The last thing is that AI will take over and my dream job would just be filled with a bunch of computer robots and will just be out of a job or maybe a career. AI could take over the whole cybersecurity field which is not a good thing because people who have been working hard will all be for nothing being everything that you learned will be replaced with AI.

Ethical Analysis

To explore these issues I decided to apply Scott Rae's moral decision making framework from Moral Choices (2018). Rare's process of identifying the moral issue, gathering the facts, evaluating options, and making morally justified decisions. In the Moral choices Rae talks about how we need to identify the ethical issues and the moral choices. The moral choice of cybersecurity using AI is that a person who is working to become a cybersecurity analyst should rely on AI tools that may make the job easier but AI also compromise privacy and fairness in the field. We already know that AI is going to be much faster at preventing cyberattacks and detecting attacks before they even happen but their reliance on keeping a vast amount of large data sets can lead to outreach and surveillance. Without the oversight of humans, AI may cause unintended harm. I think after evaluating everything that we can use AI but on certain conditions like we use AI but it has to be under human supervision to ensure ethical accountability. We should limit the AI data access to minimize privacy instruction. We should make sure that we avoid the full automation that eliminates human moral judgement.

Christian Worldwide Perspective

From a christian perspective technology is supposed to be something that is used to serve and protect others, not exploit or harm them. Matthew 22:37-39 it teaches us to love God and neighbor and emphasize respect human dignity and privacy. With AI there is no privacy, they hold that data forever and there is that 50/50 percent chance that your information will be put out to the world. Let's take a look at Micah 6:8 its believers to "act justly, love mercy, and walk humbly with your God," this message is guiding cybersecurity professions to use AI fairly and compassionately. Goodness, loyalty, self control are among the characteristics highlighted in Galatians 5:22-23 that are crucial for choosing ethical technology. This viewpoint holds that rather than controlling AI, it should be used to benefit humans. Christian are expected to act as stewards of technology to make sure that it advances justice and safety rather than damage.

ACM and IEEE Code of Ethics

The ACM code of ethics 2018 says “honesty, avoiding and respecting privacy”. Same with the IEE code of ethics 2018 it says to “hold paramount the safety, health and welfare of the public and act transparent in their work”. I feel like both of the codes follow closely with the christian moral perspective and they also are both directly applied to cybersecurity and AI ethics. The ACM is built around 4 different sections. We have professional responsibilities, leadership responsibilities, compliance, and general ethical principles. While the IEEE represents safety and accountability and fairness in technology. They both show up in their worn way but at the same time they both just work together. ACM contributes to human well being making sure that we ensure our work benefits and protects human rights. They also try to avoid harm by making sure that data doesn't get breached like how it would with something like AI and indirect effects such as bias misuse of AI. Then IEE works to be honest and realistic, wanting to make sure everyone is safe and make sure AI systems are not endangering individuals or communist over data breach. The ACM and IEE are both in place to make sure that people are safe and to help the AI understand the needs of technology and they both also help improve technology in different ways which is so amazing.

Conclusion

AI can be helpful but we have to be careful not to depend on it too much. If computers start doing all the world we will stop learning how to protect systems ourselves. Even though AI is fast and efficient it isn't always correct and those mistakes could lead to major problems. I think AI should be used to assist us, not replace us. For example it can help detect cyberattacks but humans should still make the final calls and take action. In the end using AI responsibly is all about balance without it things could easily spiral out of control. By following the christian moral principles and professional ethical standard cybersecurity experts can make sure AI is used in ways to protect people's rights and do good. The goal shouldn't just be creating smarter machines, it should be making sure technology continues to serve humanity and keep people safe.

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